TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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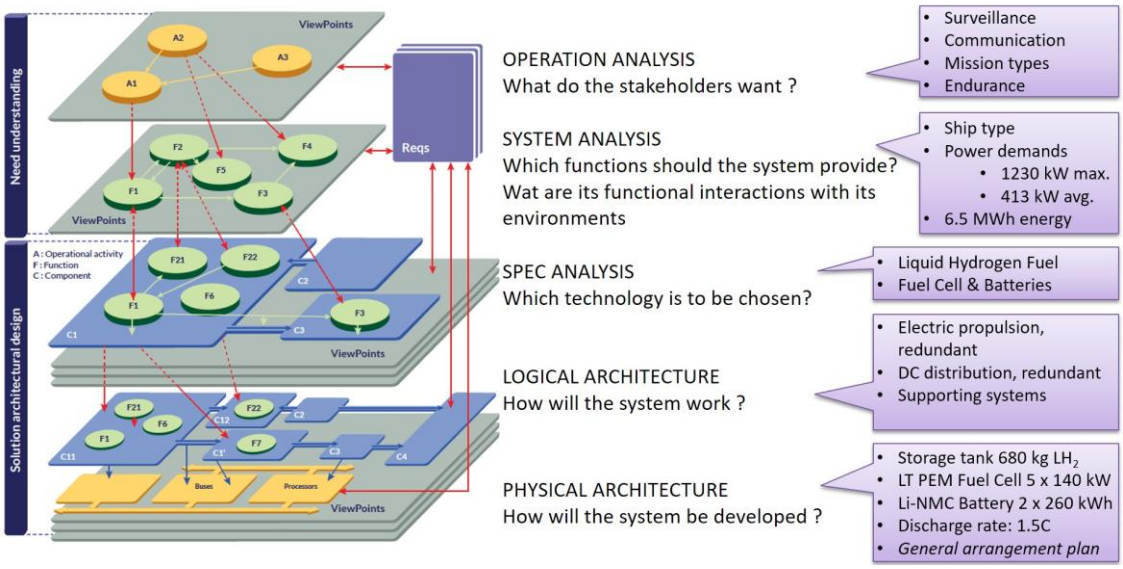
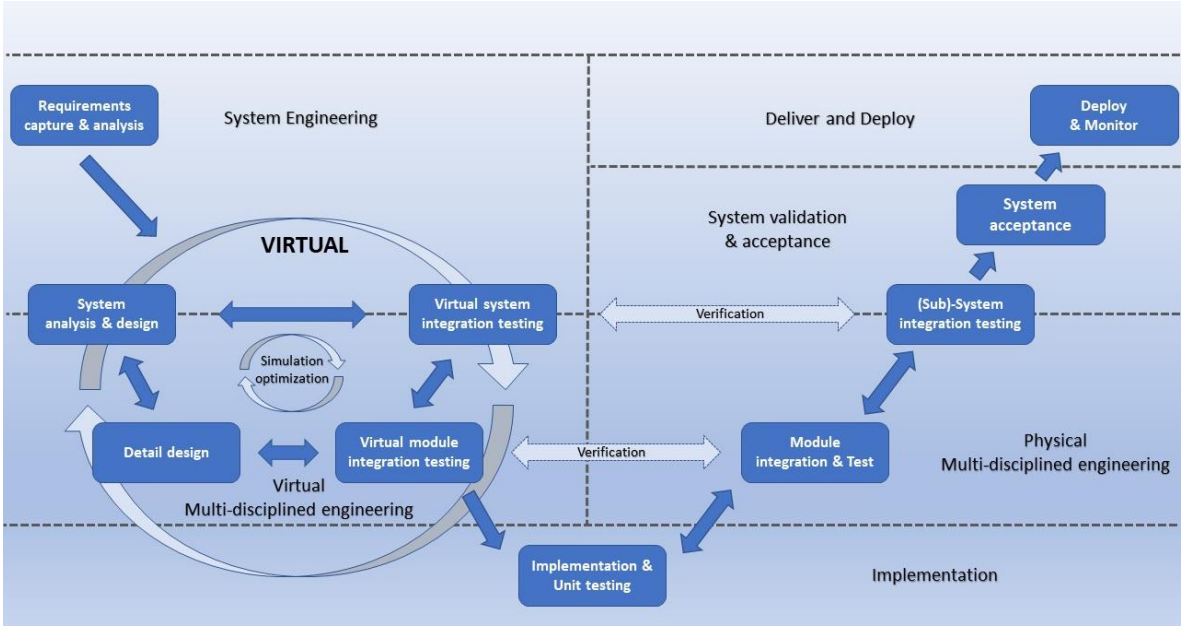
2. Technology Overview

Title of Proposed Technology	Integrated MBSE Digital Twin and Modular Energy Management Strategy (MEMS) for Vessel Design and Onboard Energy Optimisation			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (√), Next-generation Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-generation Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		**Electrical & Electronics**	
	Medium Category	Small Category	Medium Category	Small Category
	Software	SW Solutions	Batteries	Secondary Batteries
Concept of Proposed Technology	An **Integrated MBSE Digital Twin** approach combines Model Based Systems Engineering with a live digital representation of the vessel to support design, operation, and optimisation across the full lifecycle. In this framework, the MBSE model acts as the authoritative system definition, capturing requirements, functional architecture, energy flows, control logic, and safety constraints in a consistent and traceable way. When this system model is connected to onboard data during operation, it evolves into a digital twin that reflects the vessel's real-time state and behaviour.			



Applied to **vessel design**, this integration allows energy systems such as generators, batteries, power electronics, propulsion, and hotel loads to be modelled once and reused throughout development and operation. Design trade-offs, mission profiles, and operating scenarios can be evaluated early using simulation, while the same models later support validation, monitoring, and performance assessment onboard. This is especially important for electric and hybrid vessels, where power demand, environmental conditions, and operational constraints vary significantly over time.



On top of the digital twin, a **Modular Energy Management Strategy (MEMS)** provides the optimisation and control layer for onboard energy use. MEMS coordinates energy generation, storage, and consumption by selecting optimal power flows in real time while respecting system constraints such as battery state, thermal limits, safety margins, and mission objectives. Its modular structure allows the strategy to be adapted to different vessel configurations and

	energy architectures without redesigning the full control logic. Together, the Integrated MBSE Digital Twin and MEMS form a closed-loop capability. The MBSE model ensures consistency and traceability from requirements to operation, the digital twin maintains alignment between the virtual and physical vessel, and MEMS uses the twin's state estimates and predictions to optimise energy efficiency, range, safety, and system lifetime during operation. This integrated approach supports informed design decisions, robust onboard optimisation, and scalable deployment across vessel types and operational contexts. • MBSE-based vessel digital twin that maintains traceability from mission requirements to Power, Propulsion and Energy (PPE) architecture and component sizing (SysML plus simulation models) [Ref. 1]. • Coupling of the MBSE model to TNO's Modular Energy Management Strategy (MEMS): a modular optimal-control approach that decomposes the energy management problem into subsystem optimisations, enabling reuse across topologies [Ref. 2, Ref. 3]. • Automated co-design loop: explore propulsion concepts (diesel-electric, battery-hybrid, fuel cell, hydrogen or methanol engines), generate supervisory control strategies, and quantify KPIs (fuel, CO2, NOx, battery stress) across operational profiles [Ref. 1, Ref. 2]. • Validation from simulation to lab using TNO Powertrains facilities, including the Innovation Centre for Sustainable Powertrains (ICSP) and TNO's maritime engine test cell in Helmond [Ref. 4, Ref. 5]. • Integration interface for the Korean SME's ship-management software: data exchange (telemetry, mission profiles), model updates, and operational optimisation outputs. • TNO project team includes Steven Wilkins and Avedis Dadikozyan, co-authors of the MOSES paper on MBSE, co-design and energy management for vessels [Ref. 1].
Rationale for Joint R&D Project	• MOSES work by Wilkins et al. identifies MBSE, co-design, and verification and validation as key enablers to reduce risk when designing PPE systems for future vessels [Ref. 1]. • The Korean SME partner contributes vessel-domain requirements, operational profiles and software integration pathways, while TNO contributes MBSE methodology, MEMS algorithms, component libraries and validation infrastructure [Ref. 2, Ref. 4]. • Joint R&D enables a demonstrator that connects vessel-level design, modular energy management optimisation and validation on representative hardware, accelerating time-to-market for low-emission vessels [Ref. 5].
Key Words	1. MBSE 2. Digital Twin 3. MEMS 4. Vessel PPE systems 5. Hybrid-electric energy optimisation

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2)]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	• Define vessel PPE use-cases and requirements in an MBSE workflow (SysML), including verification cases and KPIs (fuel, GHG, NOx, battery usage) [Ref. 6, Ref. 7]. • Develop a vessel digital twin architecture that connects requirements, functional decomposition, physical architecture, and simulation models (FMI-based co-simulation) [Ref. 1, Ref. 8]. • Implement MEMS-based modular supervisory control and optimisation for representative vessel powertrain topologies (diesel-electric, hybrid-electric, battery, fuel-cell) [Ref. 2, Ref. 3]. • Demonstrate automated co-design: jointly optimise component sizing and control strategy for multiple mission profiles (harbour manoeuvring, transit, hotel loads) [Ref. 1]. • Validate models and controllers with hardware-in-the-loop and test-cell experiments at TNO Powertrains (ICSP/Helmond), including maritime engine capability [Ref. 4, Ref. 5]. • Deliver an integration-ready demonstrator for the SME's ship-management software: data ingestion, KPI dashboards, and recommended energy strategies.			
R&D Target Specifications	• Digital twin fidelity: PPE system model supports time-domain power flows at 1 Hz or faster and includes component efficiency maps and constraints; FMI interfaces for tool interoperability [Ref. 8]. • Optimisation scope: at least 3 propulsion configurations assessed and compared under identical mission profiles; automated controller generation per configuration [Ref. 1, Ref. 2]. • Energy-performance targets: demonstrate >=10% reduction in fuel and/or GHG intensity versus baseline rule-based strategy for selected mission profiles, with documented assumptions [Ref. 1, Ref. 7]. • Controller real-time feasibility: supervisory control executes within <=100 ms on an embedded-class target (demonstrated in HIL). • Validation: at least one closed-loop test on representative hardware (e.g., engine-generator set or electric drive) in TNO facilities; report includes uncertainty bounds and repeatability metrics [Ref. 4, Ref. 5].			
R&D Plan	Total project duration: up to 36 months (aligned with TIPA Global Cooperation R&D Program). • Work Package 1 - Requirements and MBSE model: capture stakeholder needs, define SysML model, verification cases and architecture baselines [Ref. 6]. Duration: Month 0–6 • Work Package 2 - Digital twin implementation: develop component models, integrate via FMI			

	co-simulation, calibrate with available test data [Ref. 8].Duration: Month 6–15 • Work Package 3 - MEMS adaptation for vessels: configure modular optimisation for marine PPE systems, implement constraints (battery SoC, hotel loads, emission constraints) [Ref. 1, Ref. 2]. Duration: Month 12–24 • Work Package 4 - Co-design and V&V: run design-space exploration and scenario simulations; perform V&V against requirements with traceable evidence [Ref. 1]. Duration: Month 18–30 • Work Package 5 - Experimental validation and integration: HIL/test-cell demonstrations at TNO, create integration package (APIs, documentation) for SME software deployment [Ref. 4, Ref. 5]. Duration: Month 24–36

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Lead technical work on MBSE digital-twin architecture for vessel PPE systems (requirements, architecture, traceability, V&V) based on MOSES methodology [Ref. 1]. • Provide MEMS algorithms and optimisation toolchain, including modular decomposition, constraint handling and controller code generation [Ref. 2, Ref. 3]. • Provide modelling and simulation expertise (co-simulation, FMI packaging, calibration, uncertainty analysis) and tool integration [Ref. 8]. • Provide validation facilities and experimental staff at TNO Powertrains Test Centre/ICSP in Helmond, including maritime engine test cell capability [Ref. 4, Ref. 5]. • Deliver technical documentation, training sessions and integration support for deployment in the SME's ship-management software platform. • TNO technical leads: Steven Wilkins (MBSE and vessel modelling) and Avedis Dadikozyan (MEMS and co-design optimisation) will deliver the core methodology and toolchain integration [Ref. 1, Ref. 9].
Expected R&R of Korean SME Partner	• Provide vessel design context and operational mission profiles (routes, power demand, hotel loads, environmental conditions) and acceptance criteria for KPIs. • Provide interfaces and requirements for integration into ship-management software (data ingestion, dashboards, decision-support functions). • Provide access to representative vessel telemetry or sea-trial data where possible to calibrate and validate the digital twin and optimisation outcomes. • Co-develop demonstrator scenarios and participate in design reviews, verification planning and final evaluation of results. • Support exploitation and commercialisation planning in Korea and internationally, including pilot deployments with end users.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Integrated demonstrator: an MBSE-driven vessel digital twin coupled to MEMS optimisation, enabling rapid comparison of propulsion architectures and automatically generated supervisory control strategies [Ref. 1, Ref. 2]. • Engineering productivity: reduced effort to reconfigure energy management for new vessel architectures through modularity and reuse (as argued for MEMS in TNO publications) [Ref. 2, Ref. 3]. • Operational impact: quantified reduction in fuel and GHG intensity for representative missions, aligned with IMO carbon-intensity ambition for 2030 [Ref. 7]. • Validation package: test reports and calibrated models supported by TNO facilities, enabling credible evidence for shipbuilders, operators and regulators [Ref. 4, Ref. 5]. • Commercial pathway: integration-ready software components and APIs suitable for incorporation into the Korean SME's ship ICT and optimisation offerings.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization